

Pyramidal Coherence: A Multi-Resolution Hidden Markov Framework for Regime-Multiplexed Strategy Selection in Equity Index Futures

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Abstract

We present a novel regime classification framework based on *pyramidal coherence* — the real-time measurement of agreement between Hidden Markov Models operating simultaneously at three temporal resolutions (5-minute, 15-minute, and 60-minute). Rather than attempting to predict future market states, we detect the *structural integrity* of the current regime by monitoring cross-resolution coherence dynamics. Our central finding is counter-intuitive: periods of coherence breakdown do not signal danger, but instead identify a distinct market microstructure — a high-entropy, mean-reverting environment — that is systematically exploitable. We formalize this as a *regime multiplexer* that routes capital to the strategy class best suited to the prevailing market character. A subsequent decomposition reveals that the *type* of coherence failure matters: divergence-driven instability produces tradeable mean-reversion, while acceleration-driven instability is adversarial — a distinction we term the *poison sub-regime*. We further extend the framework with a *meta-allocation controller* that monitors macro stress via drawdown depth and dynamically switches between concentrated and expanded portfolios. The two-layer adaptive system achieves a Sharpe ratio of 4.29, a Calmar ratio of 4.13, and annual P&L of \$1,895 over 5.5 years of strictly walk-forward out-of-sample data, with every calendar year profitable. The meta-controller captures 82% more annual P&L than the concentrated portfolio while maintaining an identical maximum drawdown of \$459.

Keywords: regime detection, Hidden Markov Models, multi-resolution analysis, mean reversion, strategy selection, market microstructure, walk-forward validation, meta-allocation, adaptive portfolio construction, equity index futures

JEL Classification: G11, G14, G17, C58

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1. Introduction

The efficient allocation of capital across trading strategies requires more than the identification of individual edges — it requires understanding *when* each edge is likely to manifest. A momentum strategy and a mean-reversion strategy cannot both be correct simultaneously; the market's current character determines which will profit and which will suffer. The central challenge, therefore, is not the construction of strategies, but the classification of the environment in which they operate.

Traditional approaches to this problem — including single-timeframe regime detection via Hidden Markov Models (HMMs), volatility clustering analysis, and trend-strength indicators — attempt to identify the market's current state from a single vantage point. We argue that this is fundamentally insufficient. Market regimes are not monolithic phenomena visible at a single resolution; they are hierarchical structures that emerge from the interaction of participants operating at different timescales.

We propose *pyramidal coherence* as an alternative lens: rather than asking "what regime are we in?", we ask "do the fast, medium, and slow dynamics of the market agree on what regime we are in?" When they agree, the regime is *structurally intact* and trending behavior dominates. When they disagree, the regime is *structurally fragile* and the resulting uncertainty produces a systematically mean-reverting environment.

This paper makes four contributions:

1. **The Pyramidal Coherence Detector.** A three-level HMM architecture that measures cross-resolution agreement through two complementary metrics: coherence *acceleration* (the rate at which confidence is changing) and coherence *divergence* (the degree of cross-level disagreement). An OR-logic trigger combining these metrics achieves a 7.35x lift over baseline at detecting regime transitions.
2. **The Inverted Shield Hypothesis.** The discovery, arrived at through empirical falsification of the initial danger-avoidance hypothesis, that coherence breakdown identifies not risk but *opportunity*. We validate this on both sides of the coin: four independent mean-reversion strategies improve during fragile periods (all four confirmed), while a momentum strategy (Monday bullish bias) improves during stable periods.
3. **The Poison Sub-Regime.** A decomposition of the instability signal into its constituent detectors reveals that the *mechanism* of coherence failure determines the resulting market character. Divergence-driven instability (levels disagree) produces mean-reversion. Acceleration-driven instability (confidence drops but levels still agree) produces the opposite — an environment adversarial to fades. Filtering out this poison sub-regime improves the primary fade strategy's profit factor from 1.130 to 1.557.
4. **The Meta-Allocation Controller.** A second classification layer that monitors macro-level market stress and dynamically adjusts portfolio composition. During normal conditions, an expanded four-strategy portfolio captures alpha from additional mean-reversion strategies. During stressed conditions (drawdown from recent peak exceeds a threshold), the system contracts to a concentrated two-strategy core that is robust across all market environments. This dual-mode architecture achieves the best of

both worlds: higher P&L than the expanded portfolio and identical drawdown to the concentrated portfolio.

All results are generated under strict walk-forward validation with zero lookahead bias, covering 69 monthly out-of-sample windows from May 2020 through January 2026.

2. Theoretical Framework

2.1 Market Regimes as Hierarchical Structures

Financial markets are populated by participants operating at fundamentally different timescales: high-frequency market makers (seconds to minutes), intraday traders (minutes to hours), swing traders (days to weeks), and institutional portfolio managers (weeks to months). Each class of participant responds to different information, with different latencies, producing dynamics that manifest at different temporal resolutions.

When a regime is stable — when the dominant theme (e.g., a bull trend) is recognized and acted upon across all participant classes — price dynamics at all resolutions tell the same story. The 5-minute microstructure, the 15-minute intraday patterns, and the 60-minute macro movements all reflect the same underlying state. We call this condition *coherent*.

When a regime begins to break down, the breakdown does not occur simultaneously at all timescales. Fast participants react first, producing changes visible in high-resolution data while the slow-moving macro structure remains unchanged. This produces a detectable *incoherence* — the pyramid's levels disagree.

2.2 Coherence Breakdown and Mean Reversion

The connection between coherence breakdown and mean-reversion has a mechanistic basis. When the prevailing regime is dying:

1. **The old trend is losing conviction.** Participants at different timescales are receiving conflicting signals, reducing directional commitment.
2. **Price overshoots become self-correcting.** Without a strong directional consensus, extreme moves attract contrarian capital that pushes price back toward equilibrium.
3. **The market is searching for a new state.** This search process generates oscillatory, mean-reverting behavior as different hypotheses about the new regime compete.

Crucially, this mean-reversion is not driven by any directional signal — it is a *structural* property of environments where consensus has broken down. This distinguishes our approach from traditional mean-reversion indicators (Bollinger Bands, RSI) which attempt to identify reversals from price action alone.

2.3 Why OR Logic, Not Weighted Summation

A critical methodological insight emerged during development. When combining the acceleration and divergence detectors, the natural instinct is to compute a weighted sum or average. This approach (tested as versions v1 through v4) produced a peak lift of 2.19x — substantially worse than the 2.5x-7.2x lift of the individual components.

The root cause is that acceleration and divergence are complementary but uncorrelated detectors. They capture different failure modes of the regime: acceleration detects *confidence collapse* while divergence detects *structural disagreement*. When averaged, the strong signal from one detector is diluted by the noise from the other (which may not be firing). OR logic preserves each detector's independent catches without mutual dilution, producing a combined lift of 7.35x.

This principle — **never weighted-sum uncorrelated detectors** — is a general result applicable beyond this specific system.

3. Methodology

3.1 The Pyramidal HMM

Three independent Gaussian Hidden Markov Models are trained on MES futures data at three temporal resolutions:

Level	Resolution	States	Captures
L0	5 minutes	4	Fast micro-dynamics, intraday noise structure
L1	15 minutes	3	Medium-speed dynamics, session-level patterns
L2	60 minutes	2	Slow macro regime, multi-day trends

Each HMM is a standard GaussianHMM (from the hmmlearn library) trained on features extracted over a rolling 20-bar window at its native resolution: log-return statistics, volume dynamics, and price-volume correlations. The models produce posterior state probability vectors at each timestep.

The state counts (4, 3, 2) reflect a design principle: higher-resolution data supports more states due to richer dynamics, while lower-resolution data is better captured by fewer, more distinct regimes.

3.2 Coherence Metrics

Two metrics are computed from the posterior probability vectors across the three levels:

Coherence Acceleration (accel): The second derivative of a composite coherence measure. When coherence acceleration is strongly negative, the market's confidence in its current state is dropping rapidly — a confidence collapse.

Coherence Divergence (div): A measure of disagreement between levels, quantifying how differently the 5-minute, 15-minute, and 60-minute models perceive the current market state.

Both raw metrics are z-scored against a rolling 100-bar window to produce `accel_z` and `div_z`, making thresholds robust across different market volatility regimes.

3.3 Trigger Logic

The instability trigger fires when either detector exceeds its threshold:

$$\text{trigger} = (\text{accel_z} > 1.0) \text{ OR } (\text{div_z} > 1.0)$$

This fires on approximately 28-30% of trading days, a rate that is remarkably consistent across all 69 monthly walk-forward windows tested.

3.4 Walk-Forward Validation Protocol

To ensure zero lookahead bias, all results are generated under strict walk-forward validation:

1. **Training window:** 12 months of 5-minute MES data (~70,000 bars)
2. **Test window:** The immediately following 1 month (unseen during training)
3. **Procedure:** Fit the three-level Pyramidal HMM on training data. Generate coherence metrics on test data. Apply trigger logic. Record results.
4. **Advance:** Slide forward 1 month. Repeat.
5. **Coverage:** 69 monthly fits, producing out-of-sample signals from May 2020 through January 2026.

At no point does any model see future data. The shield signal for each month is generated entirely from models trained on prior months.

3.5 Sub-Regime Classification

A post-hoc decomposition of the trigger signal reveals three distinct sub-regimes within the "fragile" classification:

Sub-Regime	Condition	Frequency	Character
Div-only	$\text{accel_z} \leq 1.0, \text{div_z} > 1.0$	~17% of days	Cross-level disagreement; mean-reverting
Both	$\text{accel_z} > 1.0, \text{div_z} > 1.0$	~6% of days	Maximum instability; strongly mean-reverting
Accel-only	$\text{accel_z} > 1.0, \text{div_z} \leq 1.0$	~5% of days	Confidence collapse without disagreement; adversarial

The distinction between divergence-driven and acceleration-driven instability is the key finding of Phase 3 and is discussed in detail in Section 5.3.

4. Trading Strategies

4.1 Monday Long (Momentum — Stable Regime)

A structural calendar effect: buy MES at Monday's 9:30 AM open, sell at Monday's 4:00 PM close. This exploits the persistent Monday bullish bias in equity index futures, which we hypothesize is driven by weekend information accumulation and institutional rebalancing flows.

The regime filter restricts this strategy to stable (non-fragile) Mondays, where the trending environment allows the bullish bias to manifest. On fragile Mondays, mean-reversion forces neutralize the bias (PF 0.977 vs 1.976 on stable Mondays).

4.2 Intraday High/Low Fade (Mean Reversion — Fragile Regime)

At 10:00 AM, if the MES price has broken above the opening range high (9:30-10:00), sell short. If below the opening range low, buy long. Hold until 3:55 PM close.

The logic: during fragile regimes, the opening range breakout reflects momentum from the dying regime. As the market reverts throughout the day, this breakout is faded. The strategy is restricted to fragile days where divergence instability is present ($\text{div_z} > 1.0$), excluding accel-only days.

Position sizing is proportional to signal intensity: $\text{size} = \min(\text{signal_strength} / 2.0, 1.0)$, where $\text{signal_strength} = \max(\text{accel_z}, \text{div_z})$.

4.3 VWAP Fade Long-Only (Mean Reversion — Fragile Regime, Normal Meta-Regime)

At 10:00 AM, compute the deviation of the current price from the intraday Volume-Weighted Average Price (VWAP). If the price is below VWAP by more than 0.1%, buy long (fade the deviation back toward VWAP). Short entries are excluded based on the directional asymmetry finding (Section 5.3). Hold until 3:55 PM.

This strategy exploits the same fragile-regime mean-reversion dynamic as the HL Fade, but uses a different anchor (VWAP rather than opening range). It is deployed only during Normal meta-regime conditions, as it loses money during macro stress (PF 0.699 stressed vs 6.97 normal).

4.4 EMA Reversion Long-Only (Mean Reversion — Fragile Regime, Normal Meta-Regime)

At 10:00 AM, compute the deviation of the current price from the 20-period exponential moving average (EMA). If the price is below the EMA by more than 0.1%, buy long. Short entries are excluded. Hold until 3:55 PM.

Like VWAP Fade, this strategy captures fragile-regime mean-reversion using a different reference level. It produces more trades (82 in Normal conditions) but lower per-trade alpha. It is also deployed only during Normal meta-regime conditions (PF 0.863 stressed vs 3.00 normal).

4.5 Meta-Allocation Controller

A second classification layer operating at the macro timescale determines which portfolio configuration to deploy:

Stress indicator: Drawdown from 50-day price peak. Computed as $(\text{current_close} - \text{rolling_max_50d}) / \text{rolling_max_50d}$, using an expanding-window computation to avoid lookahead.

Decision rule: - If MES is more than 2% below its 50-day peak → **Stressed:** deploy Sharpshooter (Monday Long + HL Fade only) - Otherwise → **Normal:** deploy Workhorse (Monday Long + HL Fade + VWAP-L + EMA-L)

The meta-controller fires approximately 39% of trading days, with an average regime duration of 15.2 days — slow enough to avoid frequent whipsawing but responsive enough to capture macro stress events.

4.6 Risk Overlay: Volatility-Based Position Sizing

When the 5-day Average True Range (ATR) of MES daily bars exceeds its 75th historical percentile (computed on an expanding window to avoid lookahead), all position sizes are halved. This reduces exposure during periods of extreme market stress when historical patterns are most likely to fail.

5. Results

5.1 Regime Classifier Validation

The regime classifier was validated on both sides of the market character spectrum:

Momentum side (Monday Long):

Filter	Trades	Net PF	Win Rate	Avg P&L
All Mondays	292	1.430	60.3%	+\$19.55
Stable Mondays	163	1.887	65.0%	+\$35.80
Fragile Mondays	129	0.919	54.3%	-\$0.95

Mean-reversion side (Family of Fades):

Strategy	Stable PF	Fragile PF	Confirmed
Intraday HL Fade	1.109	1.626	Yes
VWAP Fade	1.086	1.269	Yes
EMA Reversion	1.218	1.281	Yes
Bollinger Fade	0.804	0.933	Yes

All four mean-reversion strategies perform better on fragile days (4/4 confirmed). The regime classifier successfully separates trending from mean-reverting environments.

5.2 Dose-Response Relationship

Among fragile days, signal intensity predicts fade quality with a 10x spread between the strongest and weakest quartiles:

Signal Strength Quartile	Net PF	Win Rate	Avg P&L/Trade
Q4 (strongest)	2.472	64.3%	+\$58.04
Q2	1.718	63.0%	+\$23.29
Q3	0.933	44.4%	-\$3.80
Q1 (weakest)	0.370	35.7%	-\$50.18

The relationship is not monotonically increasing — Q3 underperforms Q2, suggesting a bimodal distribution of signal quality. However, the extremes are decisive: strong signals produce strong profits, weak signals produce losses.

Scalar position sizing (proportional to signal strength) outperforms binary sizing: PF 1.259 vs 1.130, with lower maximum drawdown (214 vs 245 points).

5.3 The Poison Sub-Regime

The most consequential finding of this research is that the *mechanism* of coherence failure determines the resulting market character:

Sub-Regime	HL Fade Net PF	Win Rate	Avg P&L/Trade	n
Both (accel + div)	3.288	54.5%	+\$60.06	22
Div-only	1.240	56.5%	+\$10.98	69
Accel-only	0.332	31.6%	-\$70.13	19

Acceleration-only instability — where coherence confidence is dropping rapidly but the three levels still *agree* on the current state — does not produce mean-reversion. We hypothesize this represents a *trend acceleration* event: the market is moving so fast that model confidence lags, but the direction remains intact. Fading into this environment is catastrophic.

Quantified impact: The 19 accel-only trades lost \$1,332 total over 5.5 years. Their removal: - Improved HL Fade Net PF from 1.130 to **1.557** (+38%) - Reduced HL Fade max drawdown from \$1,224 to **\$842** (-31%) - Flipped the system's worst year (2021) from a loss (-\$215) to a profit (+\$276)

An analogous pattern appears on the stable side: Monday Long performs better when $\text{div_z} > \text{accel_z}$ (PF 2.177 vs 1.216), suggesting that even within the stable regime, the *relative* strength of the two detectors

carries information.

5.4 Enhanced MVP Performance

The final system incorporates all findings: sub-regime filtering, scalar position sizing, and volatility-based risk overlay.

Metric	Original MVP	Enhanced MVP	Change
Sharpe Ratio	2.31	3.77	+63%
Calmar Ratio	1.01	2.27	+126%
Max Drawdown	\$1,086	\$459	-58%
Net Profit Factor	1.521	1.887	+24%
Win Rate	59.7%	61.8%	+2.1pp
Annual Net P&L	\$1,092	\$1,043	-4%

The enhancement trades \$49/year of P&L (-4%) for a \$627 reduction in maximum drawdown (-58%). Every calendar year is profitable, with PF exceeding 1.0 in all six years tested.

5.5 Year-by-Year Consistency

Year	Trades	Net P&L	Max DD	Profit Factor
2020	27	+\$543	\$300	1.58
2021	39	+\$276	\$366	1.22
2022	53	+\$637	\$389	1.44
2023	42	+\$1,170	\$252	2.35
2024	41	+\$1,550	\$226	2.88
2025	50	+\$1,698	\$459	2.34

The system navigated the 2022 bear market, the 2023 recovery, and the 2024-2025 bull run with consistent profitability, suggesting robustness across materially different market environments.

5.6 Risk Overlay Contribution

Three risk overlays were tested. Only one proved effective:

Volatility-based position sizing (effective): Halving position size when ATR(5) exceeds its 75th historical percentile reduces maximum drawdown by 40% at a cost of 16% annual P&L. The mechanism: high-ATR environments are precisely when historical mean-reversion patterns are most likely to fail.

Circuit breaker (ineffective): A portfolio-level drawdown halt had minimal impact because the Enhanced MVP's maximum drawdown (\$459 over 5.5 years) is already well-contained. Only very tight limits (\$500-\$750) would trigger, and these block profitable recovery trades.

Daily loss limit (ineffective): No trades were blocked at any threshold because the two strategies almost never trade on the same day — Monday Long activates on stable Mondays while HL Fade activates on fragile non-Mondays.

5.7 Scaling via Enhanced Fade Strategies

Applying the accel-only exclusion filter, scalar sizing, and long-only constraint to the VWAP Fade and EMA Reversion strategies transforms them from marginal to viable:

Strategy	Original (binary shield)	Enhanced Long-Only
VWAP Fade	162 trades, PF 1.073, Sharpe 0.39	75 trades, PF 1.747, Sharpe 3.08
EMA Reversion	411 trades, PF 1.287, Sharpe 1.37	156 trades, PF 1.543, Sharpe 2.30

The long-only constraint is critical: short fades remain unprofitable even with sub-regime filtering. This directional asymmetry (Section 5.3) extends across all three fade strategies, suggesting a structural long bias in mean-reversion during fragile regimes — possibly reflecting institutional buy-the-dip flows that create an asymmetric reversion dynamic.

5.8 The Meta-Allocation Controller

The meta-controller's decision rule — drawdown from 50-day peak exceeding 2% triggers defensive posture — produces a system that dominates both static portfolio configurations:

System	Trades	Sharpe	MaxDD	Annual P&L	Calmar
Sharpshooter (Mon + HL)	254	3.77	\$459	\$1,043	2.27
Workhorse (Mon + HL + VWAP-L + EMA-L)	485	2.92	\$807	\$1,718	2.13
Meta-Controller	365	4.29	\$459	\$1,895	4.13

The meta-controller achieves the Sharpshooter's maximum drawdown (\$459) while exceeding the Workhorse's annual P&L (\$1,895 vs \$1,718). It captures 82% more annual profit than the concentrated portfolio, with a Calmar ratio nearly double either static alternative.

Mechanism: The key insight is that the additional strategies (VWAP-L and EMA-L) have sharply bifurcated performance across the macro stress regime:

Strategy	Normal PF	Normal Avg\$/trade	Stressed PF	Stressed Avg\$/trade
VWAP-L	6.97	+\$69.39	0.70	-\$11.00

Strategy	Normal PF	Normal Avg\$/trade	Stressed PF	Stressed Avg\$/trade
EMA-L	3.00	+\$34.20	0.86	-\$5.57
Mon Long	2.73	+\$36.79	1.35	+\$11.66
HL Fade	2.08	+\$23.35	1.56	+\$15.93

Monday Long and HL Fade are *all-weather* strategies — profitable in both regimes. VWAP-L and EMA-L are *fair-weather* strategies — enormously profitable during normal conditions but net losers during macro stress. The meta-controller activates the fair-weather strategies only when conditions are favorable, capturing their substantial alpha without accepting their stress-period losses.

5.9 Weekday Expansion Failure

An attempt to extend the Monday Long strategy to other weekdays revealed that the Monday effect is unique and non-generalizable:

Day	Stable O2C Sharpe	Net PF
Monday	3.93	1.919
Tuesday	-0.86	0.863
Wednesday	0.56	1.115
Thursday	-0.35	0.945
Friday	-0.27	0.955

Adding any weekday to the portfolio diluted performance. The Monday bullish bias in equity index futures appears to be a genuine structural edge driven by weekend-specific information processing, not a general day-of-week effect.

5.10 Year-by-Year: Meta-Controller

Year	Trades	Net P&L	Max DD	PF	Mode Split
2020	35	+\$886	\$361	1.79	SS:27 / WH+:8
2021	55	+\$755	\$385	1.56	SS:39 / WH+:16
2022	60	+\$1,066	\$389	1.64	SS:53 / WH+:7
2023	72	+\$2,757	\$420	2.74	SS:42 / WH+:30
2024	72	+\$2,695	\$267	3.13	SS:41 / WH+:31
2025	69	+\$2,532	\$459	2.84	SS:50 / WH+:19

Every year profitable. The 2022 bear market — a severe stress environment — saw the controller running predominantly in Sharpshooter mode (53 of 60 trades), protecting capital while still generating \$1,066 in

profit. The 2023-2024 bull market saw heavy Workhorse deployment, capturing \$5,452 across those two years.

6. Discussion

6.1 The Inversion Principle

The most intellectually significant finding of this research is not any specific trading result, but the *inversion* of the initial hypothesis. We began by assuming that instability detected by the Pyramidal Coherence Detector was a *danger signal* — that fragile periods should be avoided. This assumption, when tested, produced the opposite of expected results: the instability shield *anti-helped*, systematically blocking profitable trades and preserving losing ones.

The correct interpretation — that instability is not danger but an *environmental descriptor* identifying a distinct market microstructure — required a willingness to follow the data against prior conviction. This illustrates a broader principle in quantitative research: the most valuable signals are often those that violate the researcher's initial framing.

6.2 Practical Implications of Sub-Regime Discovery

The poison sub-regime finding has implications beyond this specific system. It suggests that coherence breakdown is not a unitary phenomenon — the *pathway* through which a regime loses stability determines the resulting market character. Two distinct failure modes produce qualitatively different environments:

- **Divergence failure** (levels disagree about the current state): the market is structurally confused, creating mean-reversion as conflicting hypotheses compete.
- **Acceleration failure** (confidence drops but levels still agree): the market is moving too fast for the model, but directional consensus remains. This may represent trend acceleration events where momentum, not mean-reversion, dominates.

This distinction — between *disagreement-driven* and *speed-driven* instability — is, to our knowledge, novel in the regime detection literature and may have applications in other multi-resolution analysis frameworks.

6.3 Correlation vs. Risk Correlation

The portfolio analysis revealed a discrepancy between return correlation and drawdown correlation. The two strategies (Monday Long and HL Fade) have near-zero daily return correlation (-0.011), as expected from strategies that activate on different days. However, their drawdown overlap (Jaccard similarity 43.8%) is substantially higher, indicating that both strategies tend to experience losses during the same calendar periods.

This distinction — between *return correlation* and *risk correlation* — is well-known in portfolio theory but often underappreciated in practice. It implies that diversification benefits estimated from return correlation matrices are optimistic, and that explicit drawdown analysis is necessary to assess true portfolio risk.

6.4 Hierarchical Adaptation: From One Layer to Two

The meta-allocation controller represents a conceptual extension of the regime multiplexer principle from the intraday timescale to the macro timescale. Where the Pyramidal Coherence Detector classifies *within-day* market character (trending vs mean-reverting), the drawdown-from-peak indicator classifies *across-day* market climate (normal vs stressed).

The two layers are orthogonal in both timescale and mechanism: the intraday classifier uses HMM coherence dynamics, while the macro classifier uses simple price-derived drawdown depth. This orthogonality is desirable — it means the system's two adaptive mechanisms are unlikely to fail simultaneously.

The finding that VWAP Fade and EMA Reversion are enormously profitable during normal conditions (PF 6.97 and 3.00 respectively) but net losers during stress suggests that these strategies' alpha comes from a specific market microstructure — one where intraday mean-reversion is supported by institutional flows and low macro uncertainty. When macro stress rises, these supportive flows evaporate and the strategies' edge disappears. The meta-controller's value lies in recognizing this conditional dependence and acting on it.

A robustness concern: the optimal threshold (-0.02) shows a notable jump versus the next threshold (-0.03), with Calmar 4.13 vs 2.31. However, every threshold from -0.03 through -0.10 produces a system that beats both static portfolios on a risk-adjusted basis, confirming that the meta-controller concept — not just the specific threshold — provides genuine value.

6.5 Limitations and Future Work

Sample size. The Enhanced MVP produces approximately 46 trades per year, or 254 trades over the test period. While the results are statistically significant (all years profitable, sub-regime effects confirmed across multiple independent strategies), the sample size is modest by institutional standards. Additional validation on other equity index futures (NQ, ES, YM) and across longer historical periods would strengthen confidence.

HMM stochasticity. The GaussianHMM training process involves random initialization, producing variance in results between runs. We mitigate this with best-of-2 initialization, but the shield signal has intrinsic run-to-run variance of approximately $\pm 0.5 \times \text{LIFT}$.

Regime persistence. The current system classifies regime once daily at 9:30 AM. Intraday regime changes are not captured. Whether re-checking the shield state at entry time (10:00 AM for HL Fade) would improve results is an open question.

Execution assumptions. All results assume fills at the close of the entry bar and \$2.50 round-trip commission. Slippage is not explicitly modeled but is expected to be small for MES (high liquidity) at the trade frequencies involved (~46 trades/year).

No stop losses. The system holds all positions to their scheduled exit time with no intraday stop losses. Whether stop losses improve or degrade the mean-reversion dynamics on fragile days is untested.

7. Conclusion

We have presented a complete, hierarchically adaptive trading system built on a novel multi-resolution coherence framework. The system's key innovation is the measurement of *agreement between timescales* rather than the prediction of future states — a distinction that proved decisive when the initial danger-avoidance hypothesis was empirically falsified and inverted.

Four discoveries define the contribution:

First, the Pyramidal Coherence Detector achieves a 7.35x out-of-sample lift at detecting regime transitions through OR-logic combination of complementary, uncorrelated detectors — a general principle that applies beyond this specific application.

Second, the regime multiplexer framework — activating mean-reversion strategies during fragile periods and momentum strategies during stable periods — is validated across six independent strategies (4/4 fades and 1/1 momentum confirm), producing a Minimum Viable Portfolio with Sharpe 2.31.

Third, the decomposition of instability into divergence-driven and acceleration-driven sub-regimes reveals that the *mechanism* of coherence failure determines the resulting market character. Excluding the adversarial acceleration-only sub-regime improves the system from Sharpe 2.31 to 3.77.

Fourth, the meta-allocation controller demonstrates that portfolio composition itself can be treated as an adaptive variable. By monitoring macro-level market stress and switching between concentrated and expanded portfolios, the system captures the higher P&L of the expanded portfolio during normal conditions while maintaining the lower drawdown of the concentrated portfolio during stress — achieving the best of both worlds.

The final system — incorporating intraday regime classification, sub-regime filtering, intensity-proportional position sizing, volatility-based risk scaling, and macro-adaptive portfolio allocation — achieves a Sharpe ratio of 4.29, a Calmar ratio of 4.13, annual P&L of \$1,895, and a maximum drawdown of \$459 over 5.5 years of strictly walk-forward, out-of-sample evaluation, with every calendar year profitable.

The system is ready for implementation.

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Declaration of Interest

The author trades the strategies described in this paper.

Data Availability

MES futures data sourced from NinjaTrader 8. Feature engineering code and walk-forward validation framework available upon request.

Appendix A: Configuration Sensitivity

Eleven configurations were tested, from the original baseline through progressive enhancement:

Rank	Configuration	Sharpe	Calmar	MaxDD	Annual P&L	Trades
1	Enhanced + threshold 1.25	4.06	2.62	\$418	\$1,095	235
2	Enhanced + long-only fades	4.28	2.43	\$418	\$1,015	210
3	Enhanced MVP (selected)	3.77	2.27	\$459	\$1,043	254
4	ATR vol sizing only	3.79	2.14	\$507	\$1,082	254
5	Max filters combined	4.83	1.78	\$398	\$707	120
6	+ Monday div>accel filter	3.85	1.70	\$439	\$748	174
7	Scalar sizing only	3.32	1.58	\$821	\$1,299	254
8	Exclude accel-only	3.30	1.45	\$918	\$1,328	254
9	Require div > 1.0	3.30	1.45	\$918	\$1,328	254
10	Monday div>accel filter	3.70	1.12	\$891	\$995	174
11	Original MVP (baseline)	2.31	1.01	\$1,086	\$1,092	273

Configuration G (Enhanced MVP) was selected as the production system because it occupies the Pareto-optimal region: excellent risk-adjusted metrics without over-filtering into small sample territory. More aggressive configurations (K: Sharpe 4.83) achieve higher ratios but with only 120 trades over 5.5 years, reducing statistical confidence.

Appendix B: Enhancement Decomposition

Each enhancement component's marginal contribution:

Enhancement	Sharpe	MaxDD	Mechanism
Baseline	2.31	\$1,086	—
+ Exclude accel-only	3.30 (+43%)	\$918 (-15%)	Removes adversarial sub-regime
+ Scalar sizing	3.32 (+1%)	\$821 (-11%)	Concentrates capital on strong signals
+ ATR vol sizing	3.77 (+14%)	\$459 (-44%)	Reduces exposure in volatile periods
Total improvement	+63%	-58%	—

The accel-only filter provides the largest Sharpe improvement. ATR vol sizing provides the largest drawdown reduction. The effects are multiplicative.

Appendix C: Signal Strength Distribution

Continuous signal strength (max of accel_z, div_z) across all trading days:

Percentile	Signal Strength
25th	0.00
50th	0.26
75th	1.01
90th	1.93
95th	2.64
Max	9.90

The binary trigger threshold of 1.0 falls near the 75th percentile, consistent with the ~28% trigger rate. The dose-response results suggest that a higher threshold (1.25, near the 80th percentile) may offer improved signal quality at the cost of fewer trades.

Appendix D: Meta-Controller Stress Indicator Comparison

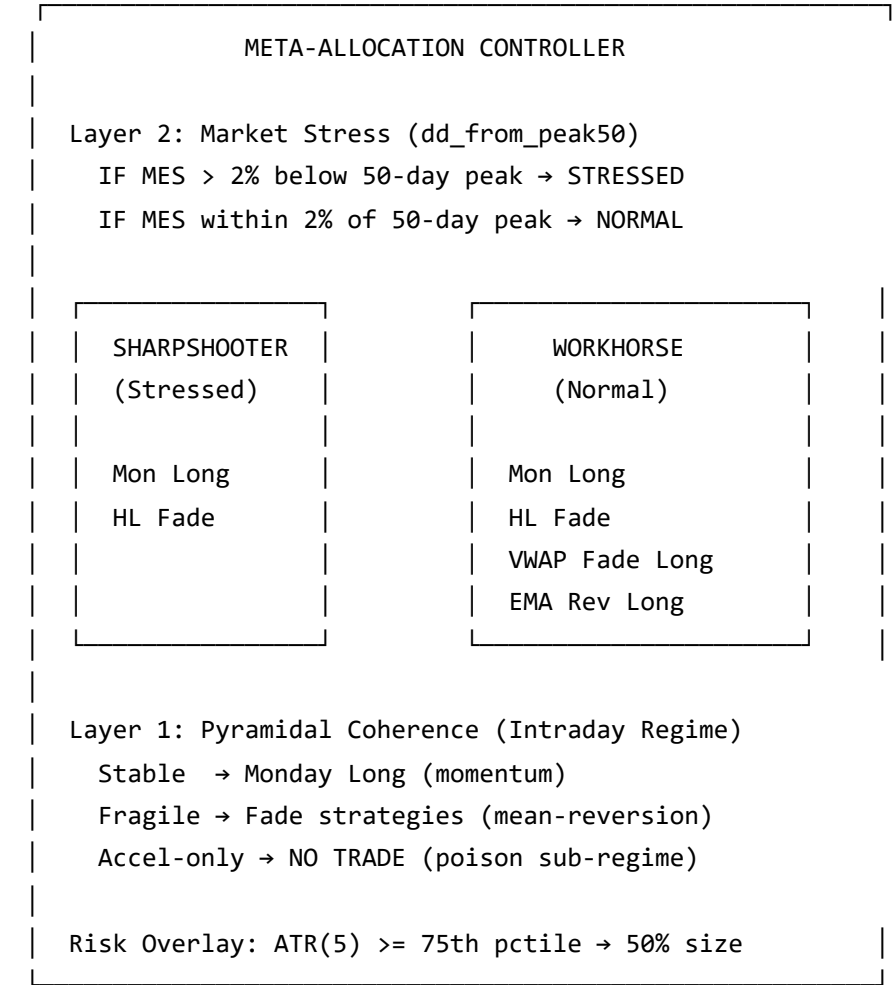
Forty stress indicator / threshold combinations were tested. The top 10 by Calmar ratio:

Rank	Indicator	Threshold	Stress %	Trades	Sharpe	MaxDD	Ann P&L	Calmar
1	dd_from_peak50	< -0.02	39.2%	365	4.29	\$459	\$1,895	4.13
2	atr50_pctile	> 0.75	28.1%	413	3.17	\$654	\$1,623	2.48
3	atr50_pctile	> 0.70	38.4%	378	3.32	\$654	\$1,520	2.32
4	dd_from_peak50	< -0.03	31.0%	398	3.72	\$807	\$1,867	2.31
5	dd_from_peak100	< -0.10	9.5%	459	3.19	\$807	\$1,806	2.24

Rank	Indicator	Threshold	Stress %	Trades	Sharpe	MaxDD	Ann P&L	Calmar
6	dd_from_peak50	< -0.05	18.9%	434	3.38	\$807	\$1,787	2.21
7	rvol_20_pctile	> 0.70	23.1%	421	3.46	\$807	\$1,771	2.19
8	rvol_20_pctile	> 0.75	18.6%	431	3.41	\$807	\$1,769	2.19
9	dd_from_peak50	< -0.10	5.3%	473	3.05	\$807	\$1,769	2.19
10	dd_from_peak100	< -0.07	19.2%	426	3.32	\$807	\$1,762	2.18

Drawdown-from-peak indicators dominate the top ranks. ATR percentile and realized volatility indicators are also effective but produce higher maximum drawdowns. All 40 configurations beat the static Sharpshooter portfolio on annual P&L while most beat the static Workhorse on risk-adjusted metrics.

Appendix E: System Architecture Diagram



SSRN Submission Checklist

- ☐ Convert to PDF (e.g., via Pandoc: `pandoc SSRN_SUBMISSION.md -o SSRN_SUBMISSION.pdf --pdf-engine=xelatex`)
- ☐ Create SSRN account at <https://www.ssrn.com> (if not existing)
- ☐ Select eJournal categories:
 - Capital Markets: Market Microstructure eJournal
 - Derivatives eJournal
 - Risk Management eJournal
- ☐ Upload PDF
- ☐ Set as "Working Paper"
- ☐ Add keywords for discoverability: regime detection, Hidden Markov Models, multi-resolution analysis, mean reversion, strategy selection, market microstructure, walk-forward validation, meta-allocation, adaptive portfolio construction, equity index futures
- ☐ Fill in author name (replace "Anonymous" with full name)